Rational Root Theorem

Rational Root Theorem: If $f(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$ has integer coefficients, then every rational zero of $f(x)$ has the following form:

$$\frac{p}{q} = \frac{\text{factors of the constant } (a_0)}{\text{factors of the leading coefficient } (a_n)}$$

Example 1: Find the rational zeros of $f(x) = x^3 - x^2 - 9x + 9$

p =

q =

Example 2: Find the rational zeros of $f(x) = x^3 - 4x^2 - 11x + 30$

Example 3: List all of the possible rational zeros of $f(x) = 15x^4 - 68x^3 - 7x^2 + 24x - 4$